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Time synchronization over a free-space optical communication channel

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Future high-precision free-space optical (FSO) clock networks will require laser links to transmit time-frequency between sites and, with this information, to synchronize the times across widely separated clocks. FSO communication networks already use laser links between remote sites to transmit high-speed data. Here we repurpose a FSO digital communication system and use it directly for two-way time-frequency transfer. We demonstrate synchronization of the time between two sites separated by a turbulent air path of 4 km using binary-phase-modulated continuous-wave laser light. Under synchronization, the two sites exhibit a fractional frequency deviation below 10^{-15} at 1500 s averaging time and a time deviation below 1 ps at averaging times of seconds to hours. Over an 8 h period, the peak-to-peak wander is 16 ps. This method should be applicable to future ground-to-space and intra-satellite links and could lead to an improved global navigation and satellite system.

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1. INTRODUCTION

Free space optical (FSO) communications systems are an appealing alternative to conventional free-space RF communications due to the higher available data rates for applications such as cellular back-haul, fiber backup during disaster recovery, metropolitan area networks, secure communications, and satellite links [1]. Here we explore the use of a digital FSO communication system for a different application, namely, to support optical clock networks via time transfer. Many applications for optical clock networks are being studied including phased array telescopes, geodesy, and precise navigation and timing [2].

In the past, we have demonstrated optical two-way time-frequency transfer (O-TWTFT) by the two-way transmission of coherent frequency comb pulse trains, combined with linear optical sampling. This comb-based O-TWTFT has shown femtosecond level synchronization over free space and is therefore capable of supporting state-of-the-art optical clocks [3–6]. However, this system is complicated, requiring three frequency combs for a point-to-point link and high-fidelity transmission of analog frequency comb pulses. Moreover, this level of timing precision is not always required, e.g., in the case of radar at low GHz carrier frequency or sub-cm positioning. In this work, we present a time transfer method over a free-space link that uses only a digital optical communication channel. Although based on the "coarse" subsystem of comb-based O-TWTFT, this work expands the timing performance significantly beyond the requirement of <2.5 ns performance of Ref. [3] to the picosecond-level by refinements of the algorithms and hardware. This approach still exploits two-way time transfer (and therefore the reciprocity of the atmosphere) but requires no additional hardware beyond the FSO communication link. The FSO link is based on a half-duplex architecture, binary phase-shift keying, and coherent heterodyne detection, but similar results should be achievable with any single-mode spatial communication link since the fundamental requirement of reciprocal time of flight is satisfied for such a link [7].

Until recently, free-space time transfer has been dominated by microwave techniques. Microwave satellite methods achieve synchronization to the nanosecond level using spread-spectrum pseudorandom modulation on a microwave carrier [4]. This performance is comparable to GPS common-view time transfer techniques [8,9]. GPS carrier phase methods have achieved performance at the tens of picoseconds level [10].

Optical-domain time transfer techniques should be able to achieve significant performance improvements over these microwave methods [2]. Indeed, fiber links allow for state-of-the-art precision time transfer [11–13] and frequency transfer [13–19] over distances ranging from hundreds of meters to hundreds of kilometers. However, these fiber-based approaches require dedicated, fixed fiber links. Thus, the development of free-space alternatives is desirable.

Several different approaches have been explored for optical time-frequency transfer over free space. The aforementioned comb-based O-TWTFT has been shown to synchronize clock times to within femtoseconds up to 12 km of air and in the presence of Doppler shifts [3–6,20,21], but at the cost of system complexity. Recent one-way phase compensation methods have reached picosecond timing wander over 100 m distances [22]. Vastly longer distances have been demonstrated with satellite-based FSO time transfer methods. Sub-nanosecond timing has been achieved by bouncing optical pulses off of nanosatellites [23]. The pulse-based time transfer by laser link (T2L2)and direct intensity-detection-based European laser timing (ELT) methods have achieved 10 ps timing wander over a day [24,25]. While

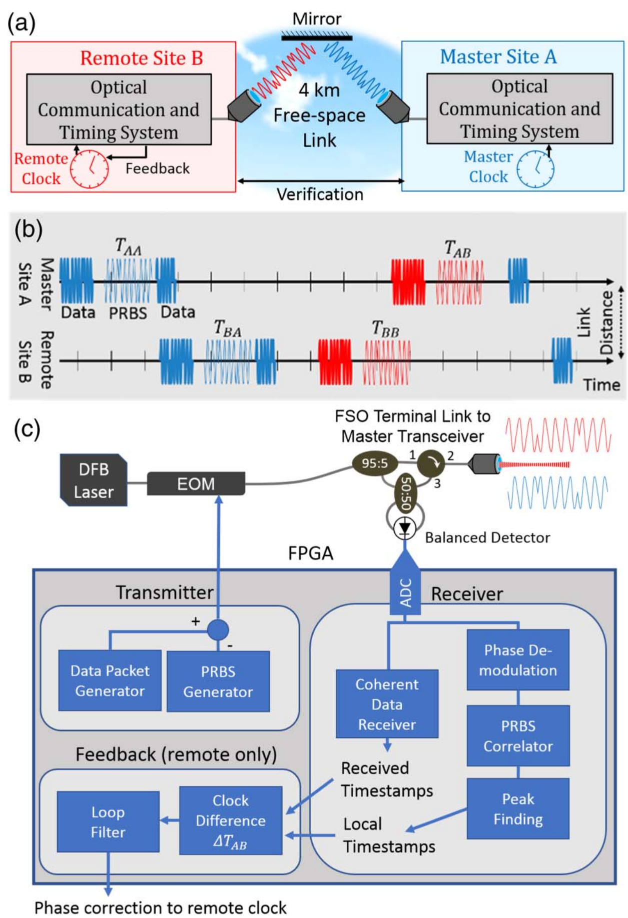

Fig. 1. (a) High-level schematic of the experimental setup. Master and remote sites are co-located because of their laboratory-based clocks and existing infrastructure [3–6], so that the link is folded via an intermediary flat mirror. (b) Diagram of the full pseudorandom binary sequence (PRBS) and data exchange on the half-duplex link. For each PRBS, two timestamps are determined, one at each site. The signals launched from the master site are shown in blue, while the ones launched from the remote site are shown in red. Data packets are time-multiplexed across the same link and are shown in bold. (c) Transceiver design for the remote site. The binary modulation is written onto the phase of the 1546 nm DFB laser through small-angle phase modulation by an EOM. FPGA, field programmable gate array; FSO terminal, free-space optical terminal; EOM, electro-optic phase modulator; DFB, distributed feedback.

these techniques have been proven to work with good performance, here we explore an alternative spread-spectrum method.

The intent here is to provide straightforward all-optical free-space time transfer and synchronization at the picosecond level that relies on existing optical communication system designs. In many ways, the approach is analogous to microwave satellite time transfer [4] and more recent work over fiber-optic telecommunication networks [26]. The system uses phase-modulated, large time-bandwidth product waveforms and coherent detection over a reciprocal single-mode link. Compared to comb-based O-TWTFT, this approach minimizes hardware overhead in systems with many stations and connections. Additionally, the phase-modulated CW laser light used here is simpler to amplify compared to comb pulses, which is important for long-distance applications such as ground-to-satellite or intra-satellite communications systems [27,28]. With this system, we show it is possible to actively synchronize a remote clock to a master clock with a time deviation (TDEV) below 1 ps from seconds to hours of averaging time and a peak-to-peak wander of 16 ps over 8 h.

Figure 1(a) shows the general layout of the synchronization system, consisting of a master and a remote site. The clock at each site consists in this case of a cavity-stabilized laser locked to a frequency comb, but could be a simple RF oscillator as well [4]. An optical coherent communication link connects the sites and is used both for O-TWTFT and for data transfer. At the remote site, the O-TWTFT timing data are used to calculate the clock offset in real time, which is then input into a phase-locked loop to actively synchronize the remote clock to the master clock. Our frequency comb-based O-TWTFT in Ref. [3] (accurate to a few femtoseconds) runs in parallel to verify the time synchronization achieved here with only the coherent communication-based system.

2. EXPERIMENTAL SETUP

We establish a digital optical coherent communication link between the sites based on binary phase-shift keyed (BPSK) modulation [29] of CW laser light. This digital communication channel is exploited for both timestamp generation and data transmission. The timestamps are transmitted via an extended waveform, comprising a pseudorandom binary sequence (PRBS) and local clock counter. The system runs in half-duplex mode, such that data can be transferred in both directions but not simultaneously.

For O-TWTFT, the system measures timestamps for four events occurring either at the master or the remote site. An example of this exchange is shown in Fig. 1(b). We first measure the transmit time, T_{AA}, of the PRBS waveform from the master site according to the local master clock. After a delay due to the link distance, we record the arrival time of this extended waveform, T_{BA}, at the remote site according to the local remote clock. The same procedure is done for a waveform sent from the remote site, yielding two other event timestamps: T_{BB} and T_{AB}, recorded according to the remote and master clocks, respectively. If this procedure is accomplished quickly enough (here in 400 µs), the transmitted waveforms both experience the same free-space turbulence, and therefore have equal link delays, due to the reciprocity of the single-mode link [6,7]. The communication link transfers the measured timestamp data from the master site to the remote site. Following the general prescription for two-way time transfer, the clock time difference is computed at the remote site as the following linear combination:

$$\Delta T_{AB} = \frac{T_{BA} - T_{AA}}{2} - \frac{T_{AB} - T_{BB}}{2} + \Delta T_{\mathrm{cal}}. \quad (1)$$

Note that ΔT_{cal} includes fixed differential delays (due to fiber paths as well as photodetector and RF components) in the transceivers.

Figure 1(c) shows the details of the transceivers used for timing exchange. At each site, a field programmable gate array (FPGA) is used for packet data communication and timestamp generation (see Supplement 1, Fig. S1). A 100 mW, 100 kHz linewidth, distributed feedback (DFB) communications laser at 1546 nm provides the optical carrier. An electro-optic phase modulator (EOM) encodes the RF waveform on the carrier phase and is driven by a digital general-purpose input/output pin of the FPGA. To shift the modulation away from the CW carrier (and associated phase noise), Manchester encoding is used. The laser light is split by a 95:5 coupler. The 95% signal is launched while the 5% tap serves as a local oscillator for the received light. The heterodyne signal is detected using a balanced detector, which allows effective detection of all received power and rejection of any intensity noise on the local oscillator. The received heterodyne signal is sampled by an analog-to-digital converter (ADC) at 200 MS/s and processed on the FPGA. This heterodyne signal is locked to ~250 MHz through feedback to the DFB laser frequency at the remote site using two actuators: an external acousto-optic modulator (AOM) for fast actuation, and the laser temperature for slow, larger range control. The error signal for this lock is computed directly in the remote site FPGA from the demodulated phase (see Supplement 1, Fig. S1). Note that the low data rate makes the use of optical quadrature demodulation unnecessary in this case, simplifying the required hardware. The entire system uses standard telecommunication components.

To recover the serialized data from the data packet generator, the coherent data receiver module in the FPGA uses several standard techniques from optical communications [29]. First, we demodulate the phase of the received heterodyne beat at ~250 MHz. The data clock is recovered first using a short training sequence at the start of each data packet, then the word boundaries are recovered using a start word, and finally the data packet contents are recovered. The data integrity is verified using a cyclic-redundancy check (CRC). No forward-error correction (FEC) was implemented, as the synchronization system can tolerate dropped packets.

To modify the communications system for time transfer, a PRBS generation system was added on the same channel in a time-multiplexed fashion. While the data packets themselves could theoretically be used for timing purposes, a known PRBS is used as the modulation type to ensure large ambiguity range and narrow pulse correlation with a fixed template. The full sequence is 1024 chips with 100 ns chip duration (10 MHz chip rate). Each PRBS is transmitted along with a local 64-bit clock counter at the transmission time, yielding for all practical purposes an infinite ambiguity range of 5 ns $\times$ 2^{64} (equivalently ~3×10^{19} m).

The demodulated phase containing the received PRBS chips is sent to a PRBS correlator at a fixed delay after a special packet announcing the transmission of a PRBS sequence. This packet allows us to reduce computational overhead by computing the correlation in a window near the estimated correlation peak time.

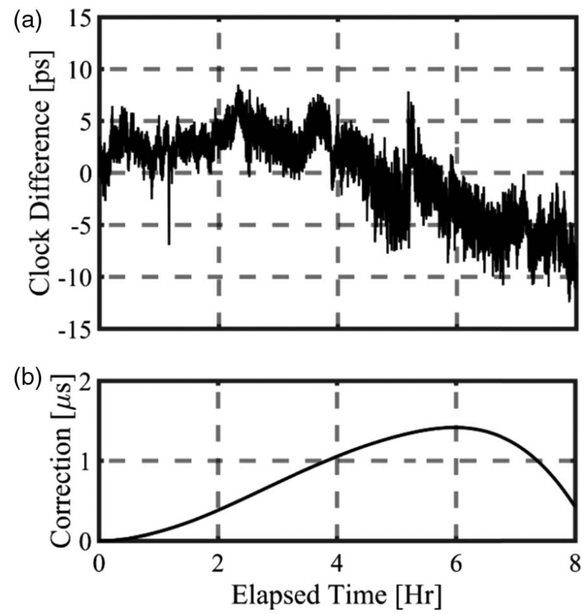

Fig. 2. (a) Clock difference at 1 s sample rate and (b) clock correction applied to the remote clock under active synchronization over approximately 4 km of air. The standard deviation is 4 ps over 8 h with a temperature-driven peak-to-peak wander of 16 ps. The system corrects for ~1 μs of timing excursion between the clocks over the measurement period, demonstrating a suppression of five orders of magnitude. Data is resampled to 1 Hz.

The resulting correlation signal (see Supplement 1, Fig. S2) has a parabolic-shaped peak whose center is estimated in real time in the FPGA, thereby generating a timestamp to subsample precision, i.e., to well below the ADC clocking period of 5 ns. During calibration, we measured a 14 ps peak-to-peak bias on the subsample delay (corresponding to a subsample bias of 14 ps/5 ns = 0.0028) that depends on the timestamp position with respect to the sample clock. We apply an approximate correction using a lookup table of the measured bias, such that any residual bias is well below the timing noise of the system.

After the master site timestamps are computed and sent over the data link to the remote site, the remote site computes Eq. (1) to find the clock difference at a 2.2 kHz update rate. This difference is sent to a Kalman filter, which provides robust estimates of ΔT_{AB} in the presence of signal dropouts [4]. The remote comb is then steered by sending the output of the Kalman filter to a simple proportional–integral loop filter on a direct digital synthesizer (DDS) mixed in with the comb stabilization lock to drive ΔT_{AB} to zero.

The bandwidth of this synchronization feedback is controlled by the dynamical response of the Kalman filter, whose optimal response time depends on the noise on the measured ΔT_{AB} versus the frequency random walk of the clock's local oscillator. Clearly, the more stable the local oscillator, the lower the optimal synchronization feedback bandwidth. Effectively, by lowering the bandwidth, we average the measured time offset until its uncertainty is below the timing drift of the local oscillator. In our case, the local oscillator is a stable optical cavity, and the Kalman filter converges to a −3 dB bandwidth of approximately 0.05 Hz to give optimal synchronized performance. If instead, the local oscillator was a quartz oscillator, then the Kalman filter settings would provide a somewhat higher synchronization bandwidth, as discussed later.

The clock synchronization is verified by a separate out-of-loop timing measurement made through the direct two-way exchange of the frequency comb light itself from each site, as in Refs. [3–6]. This out-of-loop timing measurement operates entirely independently of the communications-based (comms-based) O-TWTFT and has been independently verified to have a measurement noise floor of tens of femtoseconds [3], or well below the synchronization noise found here.

3. RESULTS

Figure 2(a) shows the time difference between the clocks during synchronization over 4 km distance, measured with the out-of-loop timing data. The peak-to-peak wander over 8 h is 16 ps with a standard deviation of 4 ps. As shown in Fig. 2(b), to achieve this level of synchronization, the remote clock time is adjusted by ~1 μs. We next analyze these data in terms of the usual metrics of power-spectral density (PSD) and Allan deviations.

The PSD for the relative timing noise between the clocks is shown in Fig. 3 for the data in Fig. 2. In addition, we show results for a shorted link under both free-running (non-synchronized) and synchronized operation. The free-running and synchronized traces cross at a location given by the Kalman filter steady-state response (lower than the synchronization bandwidth). Above the crossover, the oscillators (i.e., cavity-stabilized lasers used for this measurement) have a timing PSD lower than the synchronization system. However, at lower Fourier frequencies below the crossover, the f^{-4} relative noise of the oscillators dominates, leading to a significant clock difference if not for the active synchronization system's feedback loop. Indeed, the free-running relative timing noise is suppressed by four orders of magnitude at 1 mHz by the synchronization. The integrated timing jitter from the synchronization bandwidth to 0.2 mHz is 1.5 ps for active synchronization over the 4 km path and 300 ps for the free-running case.

The fractional frequency uncertainty given by the modified Allan deviation (MDEV) is shown for the same data in Fig. 4. At 1000 s, the synchronized clocks agree in frequency at the 10^{-15} level. The slightly higher MDEV for the 4 km data over the shorted data at averaging times beyond 1000 s is attributed to the larger temperature swings of 3 deg Celsius seen by the transceivers due to the open windows. At very short timescales, the Kalman-filter-based feedback system increases the MDEV over the free-running system. This same effect is observed in the PSDs in Fig. 3, where the synchronized PSD is higher than the free-running PSD at high frequencies. This is to be expected, as the Kalman-filter-based loop filter corrects the remote clock based on the in-loop timing measurements made via the comm-based O-TWTFT, which has a much higher measurement noise floor than the out-of-loop verification. In addition, the Kalman filter solution minimizes the unfiltered mean-squared error at each time step rather than the MDEV, which has a built-in band-pass response, peaking at $1/\tau$, for each averaging time, τ.

The synchronized clocks maintain a TDEV (Fig. 5) below 1 ps beyond 1 h. In our system, the local oscillators were cavity-stabilized lasers, which have low intrinsic phase noise. However,

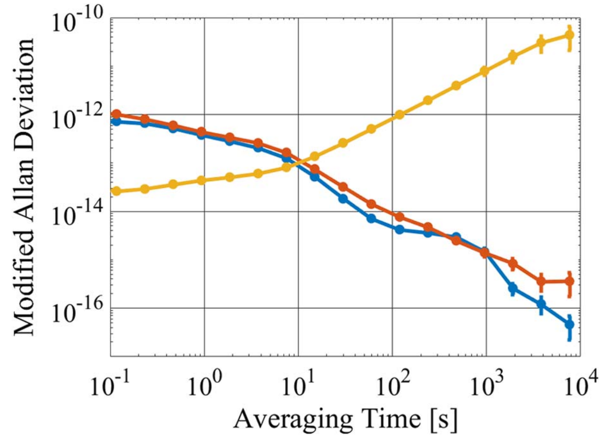

Fig. 4. Modified Allan deviation at 4 km (red) and over a shorted link (blue) under active synchronization and for a free-running system (yellow). The inverse synchronization bandwidth of the system is where the traces cross at 10 s.

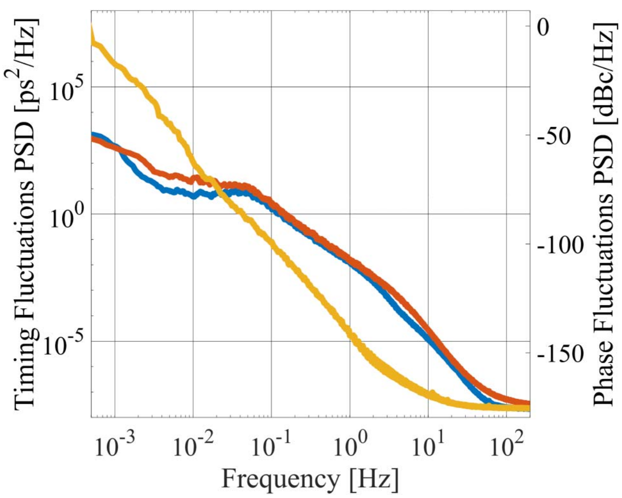

Fig. 3. Timing power spectral density of the clock time difference at 4 km (red) and over a shorted link (blue) under active synchronization and for a free-running system (yellow). Traces smoothed for visibility. The right axis shows the equivalent phase noise PSD for a nominal carrier frequency of 10 MHz.

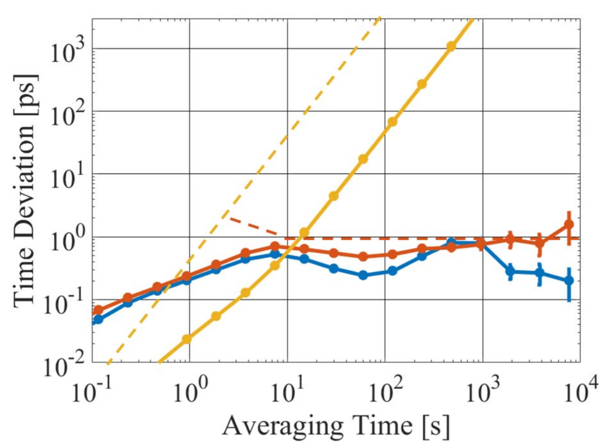

Fig. 5. Time deviation for synchronized clocks over 4 km link (red dots) and shorted link (blue dots), and time deviation for free-running unsynchronized clocks (yellow dots). The estimated free-running time deviation for remote clock based on a quartz oscillator is higher (dashed yellow line) but with synchronization should also drop to 1 ps (red dashed line) above 10 s.

a practical system might instead use a simpler quartz oscillator. Figure 5 also shows the expected free-running and synchronized performance for a remote clock based on a quartz oscillator, which reaches 1 ps for times from 10 s to 1 h as well. This estimate is based on the measured f^{-4} phase noise of a quartz oscillator [16] and optimally chosen Kalman filter parameters.

4. PERFORMANCE LIMITATIONS

Here, as in general TWTFT, we rely on the reciprocity, or equal time of flight, of the bidirectional signals transmitted over the single-mode link. However, there are four potential causes of a breakdown in this reciprocity: strong turbulence, time-varying turbulence, line-of-sight platform motion, and transverse platform motion (which leads to point-ahead effects). We briefly discuss each below, but note that both theory and experimental work with comb-based O-TWTFT establish that any reciprocity breakdown is at the femtosecond level or 1000 × below the noise floor of this comm-based O-TWTFT [3].

While increased turbulence strength will reduce the received power, it does not cause a fundamental breakdown in reciprocity. Reference [5] found, in agreement with theory [7], that reciprocity still held even over a strongly turbulent 12 km horizontal air path. However, time-varying turbulence will reduce reciprocity if the two bidirectional signals are transmitted asynchronously. However, Ref. [6] quantified this effect, finding minimal impact at the level of $\sim 10^{-17}$ fractional stability at 1 s averaging time, which is well below the stability of comm-based O-TWTFT.

Motion, both along the line of sight connecting the transceivers and transverse to their lines of sight, will cause a potential breakdown in reciprocity and is clearly an issue for future ground-to-satellite links. For line-of-sight motion, we have demonstrated that comb-based O-TWTFT can support femtosecond-level timing with closing speeds up to 25 m/s as long as the motion-induced reciprocity breakdown is properly accounted for in the algorithms [20,21]. This same approach should translate to the comm-based O-TWTFT as well, although it would need to be extended to higher speeds for a ground-to-satellite link. For transverse motion, the point-ahead effect causes a physical separation between the uplink and downlink paths, and therefore a breakdown in reciprocity. However, theoretical modeling [30,31], supported by recent ground-based tests [32], suggests that even for ground-to-satellite links, the increased timing jitter is on the order of a few to tens of femtoseconds and thus should not degrade comm-based O-TWTFT performance.

To extrapolate beyond the results here to future long-distance links, it is important to analyze the limits to the timing precision. Specifically, it is useful to distinguish between the fast noise that sets the white noise floor on the measured clock time difference and the slow noise that causes time offset drifts over minutes to hours (e.g., see Fig. 2). The former depends on the signal-to-noise ratio of the coherently demodulated PRBS sequence, while the latter is due to out-of-loop thermal drifts and communications laser carrier frequency fluctuations (which we suppressed).

Figure 6 shows the standard deviation of the measured time offset versus received power and carrier-to-noise ratio. (A common oscillator was used for both sites in this measurement, so we can attribute all timing noise to the O-TWTFT system.) For these data, the standard deviation is calculated over a short time period and thus represents the fast noise. The level of this timing noise depends on the detection SNR of the coherently demodulated PRBS (as this sets the precision with which we can define the arrival of the PRBS sequence through correlation with the known pattern).

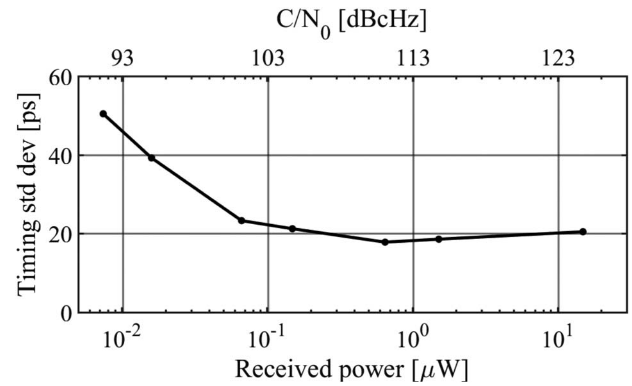

Fig. 6. Standard deviation of the measured clock time offset versus power (bottom axis) and carrier-to-noise ratio (top axis) at the 2 kHz update rate.

The detection SNR of the demodulated BPSK PRBS is limited by laser phase noise at high received powers and by additive white noise at low received powers. At powers above around 100 nW, the fast laser phase noise on the DFB lasers limits the detection SNR, giving a timing noise floor of ~20 ps (over 10 s with no averaging) independent of received optical power. Below 100 nW, detector noise and shot noise limit the SNR, so that the timing noise degrades with lower received optical power.

The location of this transition is due to the receive filter used, in this case designed to compromise between rejection of f^{-2} laser phase noise and matched filtering of the received signal (which would give optimum performance in white phase noise only). Given sufficient detection power, a different choice of communication lasers or a higher data rate could reduce the timing noise floor further. In that case, a received power of tens of nanowatts should be sufficient, which is comparable to that required/demonstrated for free-space coherent optical communications links [33–35].

The primary driver of the slow timing drift is environmental. Any noise or drift in optical or microwave delays that are not common mode to bidirectional propagation will be incorrectly attributed to clock time offset. For example, the air temperature changed by 3 deg Celsius over the measurement and this drove length changes in both RF cabling and fiber optics. Indeed, under shorted link operation, the measured timing drift is correlated to the cycling of the room temperature from the AC unit. This effect is independent of the link distance and can be reduced by tighter temperature control of the transceivers.

In addition, any wander in the frequency offset between the communication lasers will cause timing noise due to coupling of the carrier frequency and group delay. This systematic effect is a function of the group delay of the RF components in the send and receive chain. To minimize this effect, the two communication lasers are phase locked based on the measured incoming communication signal itself to a frequency offset of 247 MHz, at which value the combined group delays of the RF components is at a minimum, and the timing error is below a few picoseconds or well below the other effects discussed above.

5. CONCLUSION

We have shown a synchronization system that is compatible with an optical communications channel over a free-space link. Traversing 4 km of air, this approach yields a residual time deviation below a picosecond from 10 s to 1 h with a peak-to-peak wander of 16 ps over 8 h. The optical cavities used as local oscillators could be replaced with microwave oscillators with negligible degradation in performance.

This method provides a pragmatic tradeoff between complexity and performance, leveraging the advantages of two-way FSO transfer to provide picosecond-level timing. In systems where picosecond-level synchronization is sufficient, this approach can provide a simpler, more compact, and distance-scalable alternative to comb-based O-TWTFT. Furthermore, the picosecond timing precision is achieved in seconds, which enables real-time picosecond-level synchronization or time comparison across intermittent links.

As discussed above, extensions to longer distances and moving platforms should not result in a significant breakdown in the reciprocity on which comm-based O-TWTFT is based. As comm-based O-TWTFT is compatible with coherent FSO communications, it can be scaled up to ground-to-satellite or satellite-to-satellite distances by leveraging the advances of this more mature field to provide future ground-to-space and intra-satellite time transfer [36,37]. Optical cross-links between future navigation satellites could lead to an improved global and navigation satellite system by enabling an actively synchronized clock network. For ground-to-satellite links, e.g., this approach could provide higher timing precision than the pulse position modulation approach used by NASA's lunar laser communications demonstration [27].

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See Supplement 1 for supporting content.

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